Design and Realization of 2D Color Barcode with High Compression Ratio

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***Abstract* - With the development of barcode applied fields, the data capacity expressed by barcode is required to increase. Both enlarging the barcode size and enhancing the barcode density have some limits. We design a 2D color based on 2D barcode. This barcode is added color, and the encoded method for Chinese character is applied according to the usage frequency. Comparing with other 2D barcode, the 2D color barcode proposed in this paper has high compression ratio for Chinese character, and alpha-numeric data.**

Keywords* - *2D Color Barcode; Usage Frequency; Compression Ratio

I INTRODUCTION

Barcode technology [1,2] is mainly becoming one of the automatic recognition technologies for computer, which is widely applied in many fields, such as industry, commerce, transportation and finance. Nowadays, barcode divides into 1D barcode and 2D barcode. 1D barcode expresses the information only in horizontal direction, so the information capacity is limited. And 1D barcode can only express alphabet and figure. The information capacity of 2D barcode is 100 times of 1D barcode on the same area, and 2D barcode has many merits such as high reliability, strong capability to resist interference.

With the development of barcode applied fields, the data capacity expressed by barcode is required to increase. Under such condition, more information can be carried in unit area of barcode is required. The simplest methods to improve data capacity of barcode are to enlarge the barcode size, or enhance the barcode density. Yet such two methods have some limits. Enlarging the barcode size need to be provided with bigger printed area, yet in some actual applications, such as semiconductor chip, computer chip, or some commodity introduction with barcode, the barcode size is restricted too much to provide with enough space to contain big barcode. Furthermore, bigger barcode can affect the whole impression of commodity. Enhancing the barcode density can distinctly decrease the capability to resist interference, and can increase the difficulty of barcode recognition. We need to prepare printing equipment and recognition equipment with higher precision to solve the above issue, which leads to the high cost.

In China the 2D barcode criterions used widely are usually derived from foreign, and the foreign 2D barcode criterions are mainly designed for alpha-numeric character set. Most 2D barcode can express Chinese character, yet coding Chinese character usually uses ISN coding method, which leads to the low compression ratio and restrict the application environment of barcode.

In this paper we design and realize a kind of 2D color barcode especially aimed at Chinese character. Such color barcode has some good characteristics, such as big data capacity, high compression ratio, and good anti-jamming capability, and can be recognized rapidly and accurately.

II STRUCTURE OF 2D COLOR BARCODE

A. Color choice for 2D color barcode

Though the color barcode can improve the barcode capacity and enhance the data density, yet it may involve the segmentation of color image, which will increase the complexity of barcode recognition algorithm [3]. In order to avoid confusing the module colors, it is necessary to choose the colors with maximal contrast and otherness between each other. At the same time the selected colors must be recognised accurately, and are not easily affected by the light angle during the recognition process.

The basic method of segmentation for color image is to divide the image directly according to RGB pattern. Such method is easiest and best. For the RGB pattern, the eight colors including black, white, red, green, blue, yellow, cyan and fuchsin have big difference between each other, and can be recognized easily, so we choose these eight colors designed for the color barcode. In order to consider the encoding requirements for Chinese character, the 2D color barcode with high compression designed in this paper selects the six colors including black, white, red, green, blue and yellow. Make BarColorSet= {black, white, red, green, blue, yellow}, each value of corresponding data bit for data code is 0, 1, 2, 3, 4, 5.

B. Color sequence for 2D color barcode image

The structure of 2D color barcode is designed based on the structure of Data Matrix barcode, and the barcode image is composed of square modules arrayed regularly. The color information added not only improves the correction and recognition of image, but also makes the image more beautiful. The barcode image is divided into two parts: the first part is function region that is used to locate, correct and segment the image; the second part is data region that express the detailed data information.

The 2D color barcode image is divided into several data regions according to the amount of information. According to the amount of data regions, the 2D color barcode image can be classified into four data regions, sixteen data regions, or thirty six data regions. The function image of 2D color barcode with four data regions is shown in Figure 1, and the function image of 2D color barcode with sixteen data regions is shown in Figure 2.

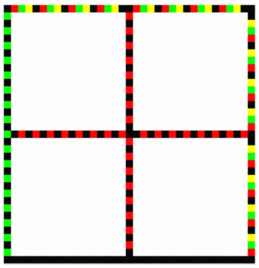


Figure 1 Function image with four data regions

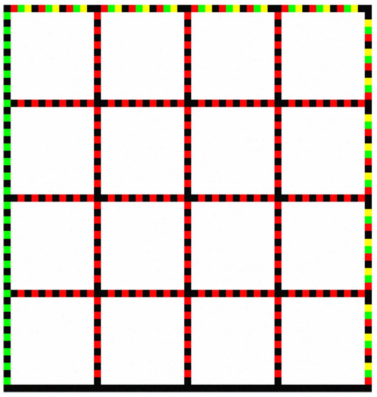


Figure 2 Function image with sixteen data regions

For the function image of 2D color barcode, the left of the image is made up of black and green modules alternated. The down of the image is composed of black modules. The up and right of image are made up of black, red, green, and yellow modules that appear regularly and repeatedly from left to right and from up to down. When the data modules overlap, it is regularly and repeatedly expressed by black, red, green and yellow modules again, which is helpful to distinguish different data regions. The overlapped module is expressed by black module. The lines that divide the data region are made up of black and red modules alternated.

III CODING FOR 2D COLOR BARCODE

A. Coded pattern for Chinese characters

Chinese characters are the main data to be encoded into barcode, and how to encode the Chinese characters is very important. There are 6763 Chinese characters and 682 graph notations in international standard character set. Yet only 3500 Chinese characters are applied in daily life, such as newspaper, broadcast, TV and network, according to the statistic and analysis by ministry of education [4]. The 600 Chinese characters used most frequently can cover 80% language samples. Table 1 shows the coverage situation for language samples. We can realize that the coverage ratio of language samples exceeds 99.2% if amount of the Chinese characters used most frequently is great than 2500. Yet if the amount is great than 3500, the coverage rate of language samples increases slowly. So the Chinese characters set encoded for 2D color barcode is designed to cover from 2500 characters to 3500 characters, because such scale has good coverage ratio and reasonable coding cost. For the other Chinese characters beyond the scale mentioned above, we can encode them according to the encoding mode of ISN. According to the calculation, the range of the code composed of five modules with five different colors can satisfy the requirements, and the range is from 0 to 3124. The coverage ratio of language samples is about 99.7%, and the corresponding value of each data bit is 0, 1, 2, 3, 4.

TABLE 1 Coverage ratio of Chinese language samples

Coverage ratio of language samples (%)	78.53	91.92	96.48	98.39	99.24	99.63	99.82	99.92
Amount of words	500	1000	1500	2000	2500	3000	3500	4000

The simplest coding mode of Chinese character database is to code each Chinese character into quinary 5-bit code. Though such method is simple and can also improve the compression ratio, yet it is not the best method, because it does not consider the situation that the increasing speed of the coverage ratio of language samples change from fast to slowly according to the increase of the word amount. According to statistic and analysis, the 625 Chinese characters used most frequently only occupy 25% of the coded character set, yet the coverage ratio of language samples is great than 80%. The other 2500 Chinese characters occupy the 75% of the coding database, yet the coverage ratio of language samples is less than 20%. It is similar as the issue of instruction set that only 20% of instructions are used for 80% of programs, so we can take the same measure as instruction set refined for coding. We design a policy that make the coded length of Chinese characters frequently used short, and make the coded length of Chinese characters seldom used long.

According to the coded database for Chinese characters established by Tsinghua University, Chinese characters are arranged according to the usage frequency from high to low. The coded values for the 600 Chinese characters used most frequently are designed from 0 to 599. And the coded values for the 25 punctuations commonly used are designed from 600 to 624. And the coding values are translated into quinary 4-bit code. The coded values of the 2500 Chinese characters are designed from 625 to 3124. The coded values are translated into quinary 5-bit code, and the first bit is added by 5, which is used to distinguish the coded length of

four bits from that of five bits.

If we encode by the usage frequency, the average length for each Chinese character code is, 4*80%+6*20%=4.4 bits. If we directly code by the ISN code, we need two bytes, 16 bits, to express each Chinese character. Even if we use the modules with six colors to encode the Chinese character, the coding length is 7 bits. If each Chinese character is expressed with the same length, and encoded by six colors, the coding length of each Chinese character is still 5 bits. So the coding method mentioned above is good enough to improve the compress ratio greatly.

B. Coded pattern of ISN

The coded pattern of ISN is a kind of complementary mean for coded pattern for Chinese character. We use the coded pattern of ISN to code the Chinese characters which are applied seldom and aren't included in the Chinese character database. The ISN is expressed by two bytes, and the value range is from 8140 to FEFE. The value range of the first byte is from 81 to FE, and the value range of the second byte is from 40 to FE. The coded policy for ISN is: the value of the first byte * 162 + the value of the second byte. And the value obtained through the above method is translated into quinary 7-bit code. The changed value range is from 20776 to 65278, and the corresponding quinary coded value range is from 1131101 to 4042103. Add 50 ahead of each code as a notation, in this way the coded length is 9 bits.

Though each Chinese character can be encoded through ISN pattern, yet the average coded length of each Chinese character is 9 bits, the compression ratio is low. The coded pattern of ISN designed in this paper is only a complementary mean to encode some Chinese characters with low usage frequency and some punctuation.

C. Coded pattern for single ASCII

There are 128 codes for ASCII, and the maximal code is expressed by quinary 4-bit code. Use the coded value of each ASCII as the coded value of barcode, and the range of coded value expressed by quinary code is from 0000 to 1002. The detailed coded policy is described as follows. Translate each coded value into corresponding quinary code. If the quinary code is less than 4 bits, we need to add 0 as the highest bit and add 50 as pattern notation. The coded length of single ASCII code is six bits.

Though the length of single ASCII code need six bits to express, such method can make the code pattern change automatically from Chinese character coded pattern to single ASCII coded pattern without switching.

D. Coded pattern for numeric data

If we apply single ASCII coded pattern to encode continuous numeric data, the coded efficiency is very low, so we design a coded pattern especially for continuous numeric data. The coded policy is described as follows. Two bits of the inputted data are considered as a group. Translate each group into quinary 3-bit code. If the quinary code is less than three bits, we need to add 0 as the highest bit. If the number of the inputted data is not the multiple of 2, the last number is translated into quinary 2-bit code. If the quinary code is less than two bits, we need to add 0 as the highest bit. Combine all the quinary code, add 515 as the notation of coded pattern, and add 5 as the ending notation.

This coded pattern is suitable for continuous numeric data, and it needs four bits for indication and three bits to code two continuous numbers. If we adopt such method, we only need 1.5 bits to encode a digit.

Ⅳ CREATION OF THE CORRECTING CODE

We take the Reed-Solomon algorithm [5,6, 7] as the correcting algorithm for the 2D color barcode and define q=1291 for GF(q). When the number of coded bit words is less than 1291, we directly take Reed-Solomon algorithm to correct code. If the number of coded bit words is great than 1291, the 2D color barcode must be divided using the Data Matrix method. We use the cross – divided method because we want to put the wrong coded word equably on the correcting sequence of each module when the barcode image can't be recognised. If the wrong part of the barcode image is centralized on the correcting sequence of one module, such module may not be recognized correctly, which also will make the whole barcode unrecognised. The correcting code function is expressed as (1).

$$g(x)=(x-2)(x-2^2)(x-2^3)\ldots(x-2^k) \quad (1)$$

Create each coded word by four bits from the dada bit. All the coded words compose the original data multinomial expressed as $d(x)$: $d_{n-1},d_{n-2},\ldots,d_1,d_0$, where d_{n-1} is the first coded word, d_0 is the last coded word. And d(x) is expressed as (2).

$$d(x)=d_0x^k+d_1x^{k+1}+d_2x^{k+2}+\cdots+d_{n-1}x^{n+k-1} \quad (2)$$

According to the correcting code algorithm shown as (3)

$$e(x)=d(x)\bmod g(x) \quad (3)$$

We can get the correcting code multinomial shown in (4). Where $e_{k-1},e_{k-2},\ldots e_1,e_0$ is correcting code, e_{k-1} is the first correcting code, e_0 is the last correcting code.

$$e(x)=e_{k-1}x^{k-1}+e_{k-2}x^{k-2}+\ldots+e_1x+e_0 \quad (4)$$

Ⅴ REALIZATION AND PERFORMANCE ANALYSIS

According to the coded rule, we encode some information including about 140 Chinese characters.. We choose 35×35 as the code specification, and create the correcting code word into the data code word. The code image created is shown in Figure 3.

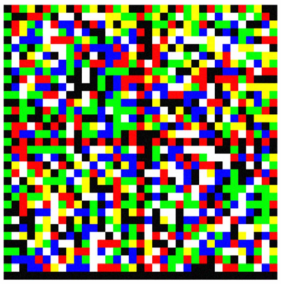


Figure 3 The code image

According to analysis, we can draw a conclusion that the performances of 2D color barcode are good that are described as follows.

(1) Information capacity

Add color information into barcode can improve the information capacity greatly, even if any coded rules are not changed. The 2D barcode designed in this paper not only take advantage of the color information, but also adjust the coded rule, which enhance the information capacity. The maximal information capacity of 2D color barcode is 3 times of that of Data Matrix or QR.

(2) Information compression ratio

The 2D color barcode has high compression ratio for Chinese characters. It averagely needs 4.4 bits to express a Chinese character. Compare to Data Matrix code or QR code, it improve 3 times. The compression ration for alpha-numeric data improved 1 time.

(3)Correcting capability

The correcting capability of 2D color barcode is improved when the size of barcode is enlarged, because 2D-barcode has big GF region that can correct many coded word. The 2D barcode has better correcting capability, and can check and correct wrong code when some coded word can not be recognized

Ⅵ CONCLUSION

In this paper we propose a coded method for Chinese character according to the usage frequency, and design the 2D color barcode with high compression ratio for Chinese character, alpha-numeric data.

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